

Shankhanil Borthakur

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EDUCATION

University of Michigan, Ann Arbor

M.S. in Computer Science and Engineering

GPA: (expected)

Aug 2026 - May 2028

National Institute of Technology, Surat

B.Tech. in Computer Science (Minor), Electronics and Communication (Major)

GPA: 3.95/4

Aug 2019 - May 2023

PROFESSIONAL EXPERIENCE

Software Engineer 2, JPMorgan Chase

Feb 2025 - June 2026

Software Engineer, JPMorgan Chase

Aug 2023 - Jan 2025

Software Engineer Intern, JPMorgan Chase

May 2022 - July 2022

- Architected a scalable data platform on AWS (Redshift, S3, Lambda, EC2) contributing to a **\$2B** foreign-exchange (FX) business.
- Deployed an event-driven pipeline that processes **5M+** daily trades with **dual-write consistency** to Redshift and Redis cache. Achieved **3X** higher **write throughput** over single-node INSERTs using Redshift COPY commands for parallel data loading.
- Reduced cloud storage spends by 30% (~\$150K annually) by designing a **federated strategy** that offloaded 70TB+ of historical trade data to Amazon S3 with **Redshift Spectrum** for on-demand querying.
- Designed a **bi-temporal data model**, enabling point-in-time reconstruction of trade state across **10+ years** of history.
- Led the incident response** for database failover, achieving **zero RPO** and **<1 hour RTO** during **2** regional cloud-service disruptions.
- Developed a **RAG agent** for AI-assisted analysis of Redshift distkey/sortkey configurations and vacuum/analyze schedules. Improved runtime of critical analytical queries by **25%**.

RESEARCH EXPERIENCE & PUBLICATIONS

C = Conference

Research Fellowship, IIT Delhi - Implemented **Twin Neural Networks** in PyTorch to address class imbalance in COVID-19 datasets (fewer positive samples) for chest X-ray classification, improving accuracy by 5%.

Indian Space Research Organization - Built a **geolocation algorithm** that maps Mask R-CNN instance segmentation outputs to GPS coordinates in satellite imagery, achieving **sub-meter positional accuracy** for remote sensing applications.

[C.1] NER for Extracting Tourist Attractions from Textual Information: A BERT-BiLSTM-CRF Approach, 2023 IEEE 5th ICAECC. (4 Cites)

[C.2] Object Detection for Military Surveillance using YOLO Framework, 2023 IEEE 20th INDICON. (11 Cites)

PROJECTS

Streak Fit | AI-Powered Virtual Fitness Platform [\[Code\]](#) [🔗](#)

- Led a **3-member team** to build a multi-user fitness platform with **WebRTC** video conferencing and **MediaPipe** pose detection, delivering real-time form-correction feedback at **30 FPS** with **4** concurrent users.

Q&A Made Easy | Real-Time Q&A Moderation System using NLP [\[Code\]](#) [🔗](#)

- Chrome extension with integrated speech-to-text transcription and an **NLP classifier** that filters **profane** and **duplicate** questions during live meetings. Automated delivery of filtered questions to the meeting chat via **Selenium**.

Travelo | Smart Travel Planning using Named Entity Recognition [\[Code\]](#) [🔗](#)

- Web application that parses travel blogs using a custom **BERT NER** model to extract tourist destinations, then generates a multi-stop itinerary with optimal **route visualization**, reducing manual trip planning time from hours to under 5 mins.

TECHNICAL SKILLS

Programming: Python, C++, Java, SQL

Tools & Technologies: Django, Flask, React, Docker, Kubernetes, REST APIs, CI/CD, Git

Cloud & Data: Terraform, AWS, PostgreSQL, MongoDB, Redshift

ML / AI: LLMs, Agentic AI, RAG, Reinforcement Learning, NLP, OpenCV, TensorFlow, PyTorch, NumPy, Pandas

CS Fundamentals: Data Structures, Algorithms, OOPs, Distributed Systems, DBMS, Operating Systems, Networking, Statistics

LEADERSHIP & AWARDS

- Gold Medalist**, ranked 1st out of 156 students in the Department of Electronics Engineering, NIT Surat.
- Accelerated Promotion** to Software Engineer 2 at JPMorgan, recognized as a **top 5%** performer in global early-engineering cohort.
- Winner** of **7** hackathon awards, including annual competitions at the **University at Buffalo** and **Fordham University**.
- Microsoft Student Ambassador:** Mentored **50+** students in web development and organized **10+** programming workshops.
- Tech for Social Good Volunteer:** Drove product development of an automation tool for a **non-profit** client with a team of 6 engineers. Supported technology transfer for users with minimal programming experience for long-term **self-sufficiency**.